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Recommender Systems

Non-Personalized Recommender Systems

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Non-Personalized Recommendations

- Do not require any information on the user.
- Recommendation does not depend on the user, but only on the items.
- Ranking is the same for all users.

Why do we need Non-Personalized Recommender Systems?

- Most platforms start with non-personalized recommendations as a cold-start fallback
- Often also used for new users (cold-start scenario) until enough data is available
- Often used as strong baseline for the evaluation of recommender systems
- Reveals structural data properties

Levels of Personalization

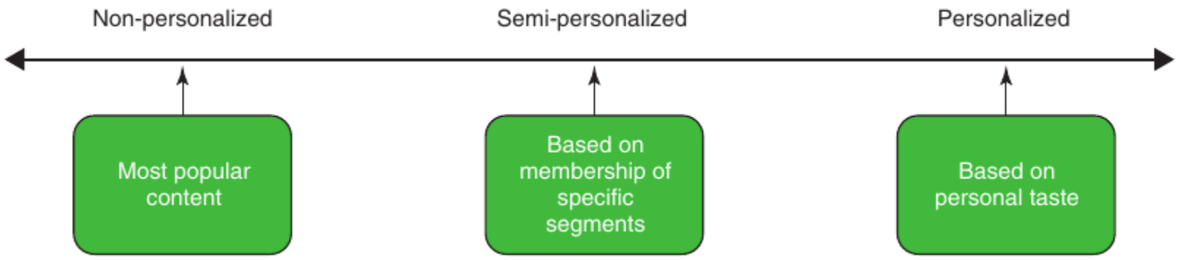


Figure taken from Practical Recommender Systems: Kim Falk; Manning Publications, 2019.

Recommendation vs. Commercial

- Commercial aims at showing customers items and convincing them that those are good deals vendor-driven
- Recommendations are always based on a calculation
- Calculation is always based on usage data data-driven

Most Popular Approach

- Show most popular items to users
- Long-tail distribution: most people will like at least some of the items in the list
- Assumption: users are similar in their preferences, so we can recommend the same items to everyone.

Most Popular Approach cont.

- Rank items by interaction frequency:

$$s(i) = \text{count}(i)$$

- Essentially aggregates demand for items across all users.
- Often used as baseline for personalized recommender systems

Popularity Bias and the Rich-Get-Richer Effect

Popular items get more popular, while unpopular items get less popular (popularity bias)

Most Popular Approach cont.

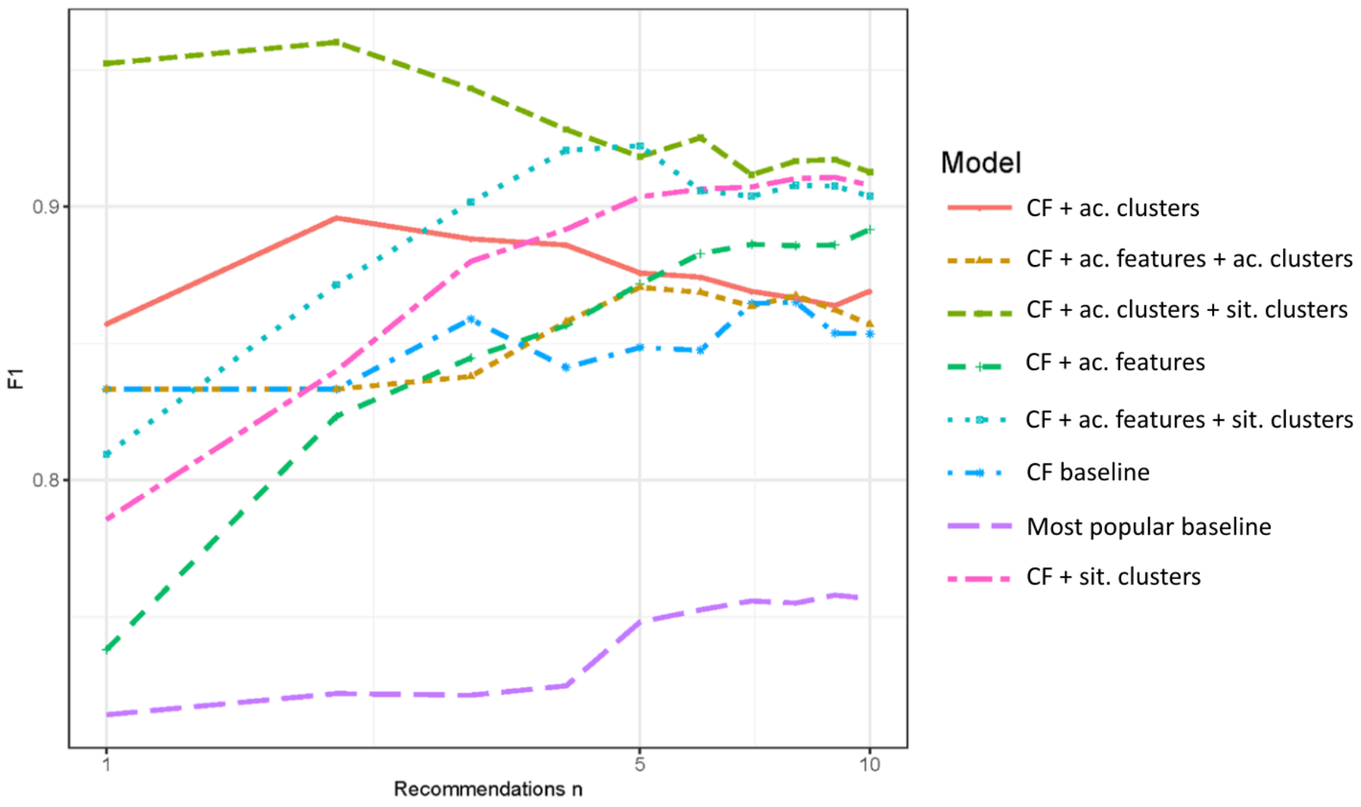


Fig. 3 F_1 for $n = 1 \ldots 10$ recommendations

Pichl, M., Zangerle, E. User models for multi-context-aware music recommendation. Multimed Tools Appl 80, 22509–22531 (2021). https://doi.org/10.1007/s11042-020-09890-7

Means Model

Simplest non-personalized prediction of ratings is to predict the mean rating for each item:

$$\hat{r_i} = \bar{r_i} = \frac{\sum_{r_{ui} \in \mathcal{R}} r_{ui}}{|\mathcal{R}_i|}$$

Time-Decay Approach

- Extends Most Popular Approach, giving recent interactions more weight.
- Models temporal shifts in demand and popularity.
- Assumption: recent demand is more indicative of near-future demand.

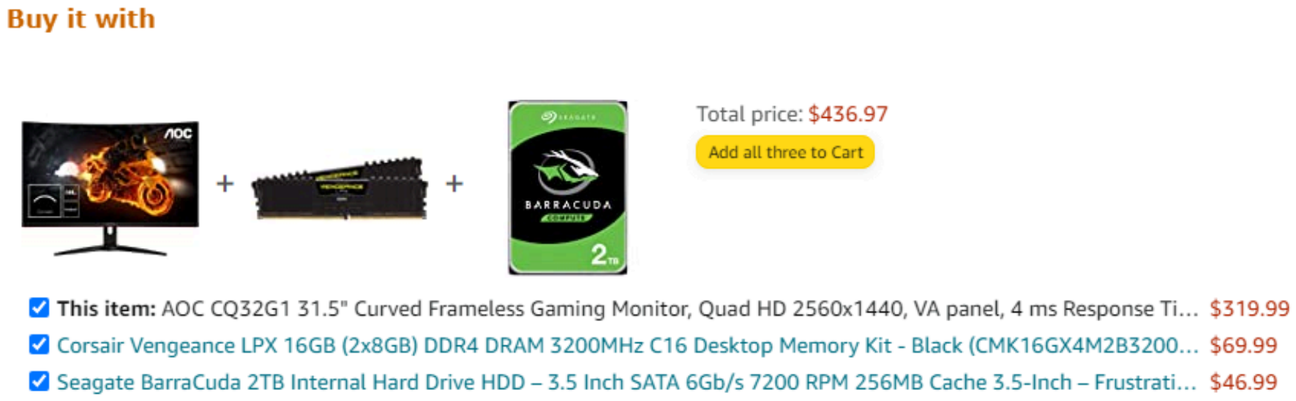
$$s(i) = \sum_t \alpha^{(T-t)} \cdot \mathbf{1}_{\text{interaction}}$$

with

- $\mathcal{D}_i$: set of interactions with item i
- t: timestamp of interaction
- T: current time
- $\alpha \in (0, 1)$: decay factor

Contextualization: Seeded Recommendations

- Contextualize recommendations to the items the user is browsing
- Association rules (for instance, via Apriori algorithm)
- Analysis of items that are frequently bought together



Rakesh Agrawal, Tomasz Imieliński, and Arun Swami. 1993. Mining association rules between sets of items in large databases. In Proceedings of the 1993 ACM SIGMOD international conference on Management of data (SIGMOD '93). Association for Computing Machinery, New York, NY, USA, 207–216.

Association Rules

- Association Rules: If-then rules about the contents of baskets
- $I \to J$, where I and J are item sets
- $\{i_1, i_2, \ldots, i_k\} \to j$ means: "if a basket contains all of $i_1, \ldots, i_k$ then it is likely to contain j"
- In practice there are many rules, we want to find interesting ones.

Association Rules: Example

- $T_1 = \{bread, yogurt\}$
- $T_2 = \{milk, bread, carrots\}$
- $T_3 = \{bread, carrots\}$
- $T_4 = \{bread, milk\}$
- $T_5 = \{milk, dates, carrots\}$
- $T_6 = \{milk, dates, yogurt, bread\}$

Association Rules cont.

- Interestingness of a rule
- Milk is frequently bought, added value of rule is limited.
- Find products that are often bought together, but not with every other item.
- Milk and bread are frequently co-purchased (3 out of 6 transactions). ➜ We could be confident that this is a sensible rule.
- Measures: support and confidence

Association Rules cont.

Support captures how often the rule appears in the data (i.e., how often the items in the rule are co-purchased).

$$support(X \rightarrow Y) = \frac{|T(X \wedge Y)|}{|T()|}$$

where $T(X)$ is the set of transactions that contain X and $T()$ is the set of all transactions.

Examples:

- $s(bread \rightarrow milk) = \frac{3}{6}$
- $s(dates \rightarrow carrots) = \frac{1}{6}$

Association Rules cont.

Confidence captures how often the rule is correct when it applies (i.e., how often the items in the rule are co-purchased given that the antecedent is purchased).

$$confidence(X \rightarrow Y) = \frac{|T(X \wedge Y)|}{|T(X)|}$$

where $T(X)$ is the set of transactions that contain X and $T(X \wedge Y)$ is the set of transactions that contain both X and Y.

Association Rules cont.

Examples:

- $c(bread \rightarrow milk) = \frac{T(bread \wedge milk)}{T(bread)} = \frac{3}{5}$ (bread appears in 5 transactions, and 3 of them also contain milk)

  ➔ 60% confidence that if a customer buys bread, they will also buy milk.
- $c(dates \rightarrow carrots) = \frac{T(dates \wedge carrots)}{T(dates)} = \frac{1}{2}$ (dates appear in 2 transactions, and one also contains carrots)

  ➔ 50% confidence, less data to support the claim

Association Rules

- Not all high-confidence rules are interesting (e.g., $bread \rightarrow milk$ is not very interesting if milk is bought in 90% of transactions).
- We want to find rules that are both frequent and have high confidence, but also provide new insights (e.g., $dates \rightarrow carrots$ might be more interesting if dates are not very common, but when they are bought, carrots are often bought as well).
- We can use combined measures like lift to find rules that are more interesting than what would be expected by chance:

$$lift(X \rightarrow Y) = \frac{confidence(X \rightarrow Y)}{support(Y)}$$

Association Rules cont.

- Step 1: Find all frequent itemsets I
- Step 2: Rule generation
  - For every subset A of I, generate a rule $A \rightarrow I \setminus A$
  - Since I is frequent, A is also frequent
  - Option 1: Single pass to compute the rule confidence
    $confidence(A, B \rightarrow C, D) = \frac{support(A,B,C,D)}{support(A,B)}$
  - Option 2:
    - Observation: If $A, B, C \rightarrow D$ is below confidence, so is $A, B \rightarrow C, D$
    - Can generate "bigger" rules from smaller ones!
  - Output the rules above the confidence threshold

Association Rules: Frequent Itemsets

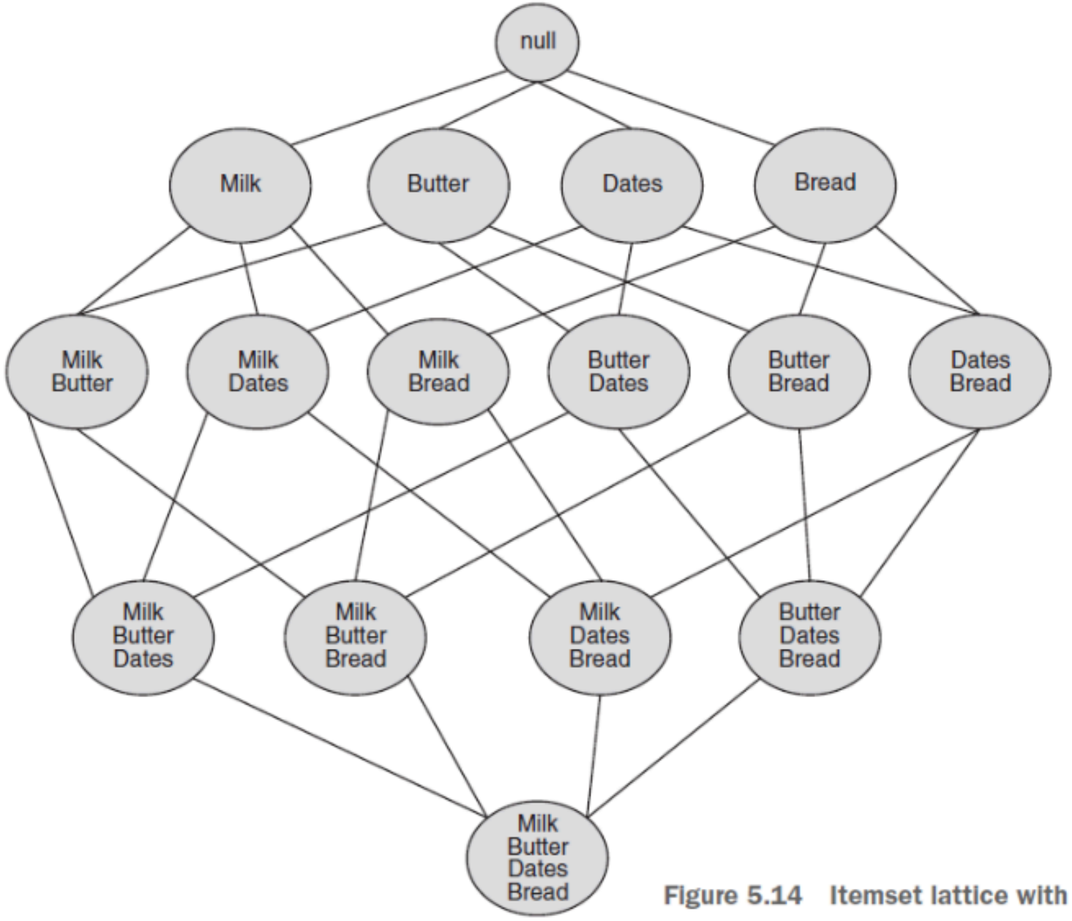
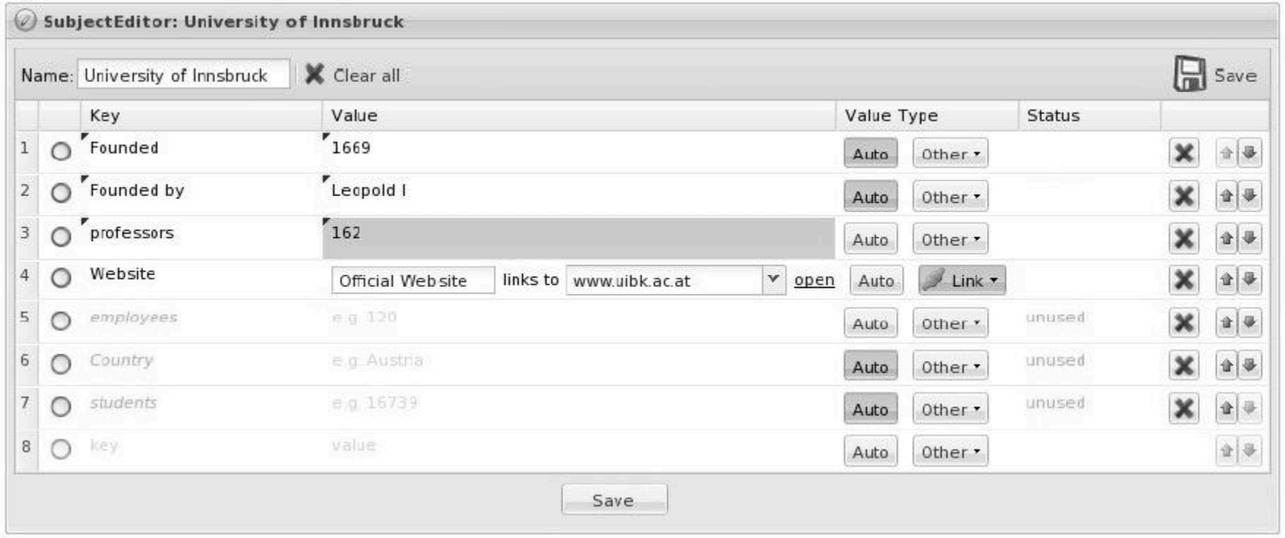
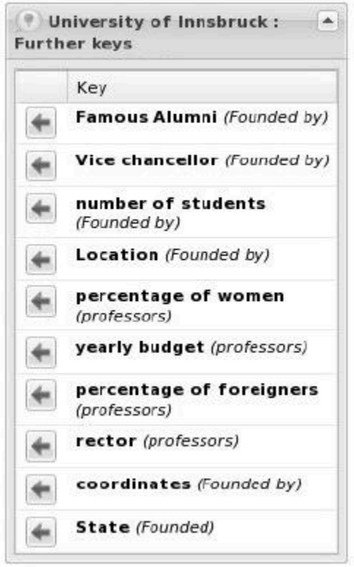


Figure 5.14 Itemset lattice with four elements

Figure taken from Practical Recommender Systems: Kim Falk; Manning Publications, 2019.

Contextualization: Seeded Recommendations





Eva Zangerle, Wolfgang Gassler, Günther Specht. Recommending Structure in Collaborative Semistructured Information Systems. RecSys '10: Proceedings of the third ACM Conference on Recommender Systems, 2010.
Wolfgang Gassler, Eva Zangerle, Günther Specht. The Snoopy Concept: Fighting Heterogeneity in Semistructured and Collaborative Information Systems by using Recommendations. The 2011 International Conference on Collaboration Technologies and Systems (CTS 2011), 2011.

Contextualization: Seeded Recommendations

Algorithm 1: Recommendation Candidate Computation

```
Input: P_{S_i}, R
Output: set C of all recommendation candidates for S_i
C ← ∅
T ← ∅
foreach p_i ∈ P_{S_i} do
    foreach (p_a, p_b, c) ∈ R do
        if (p_a == p_i ∧ p_b ∉ P_{S_i}) then
            T ← T ∪ {(p_a, p_b, c)}
        end
    end
end
foreach (p_a, p_b, c_i) ∈ T do
    if (∃(p_x, c) ∈ C, where p_x == p_b) then
        C = C \ {(p_x, c)}
        C = C ∪ {(p_x, c + c_i)}
    else
        C = C ∪ {(p_b, c_i)}
    end
end
return C
```

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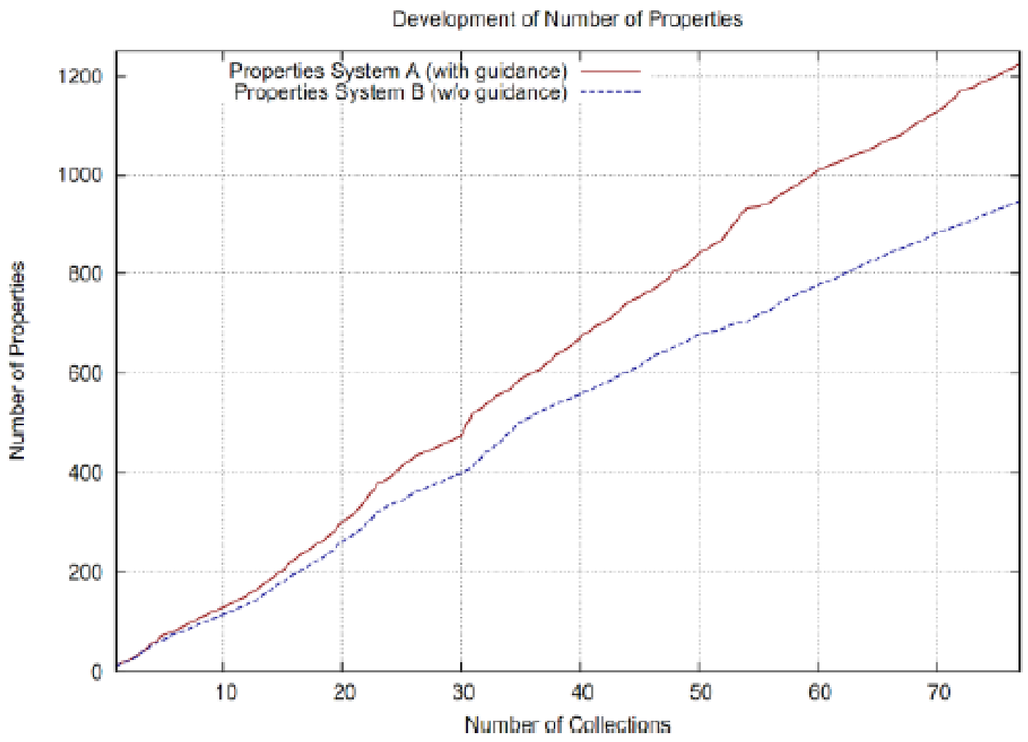

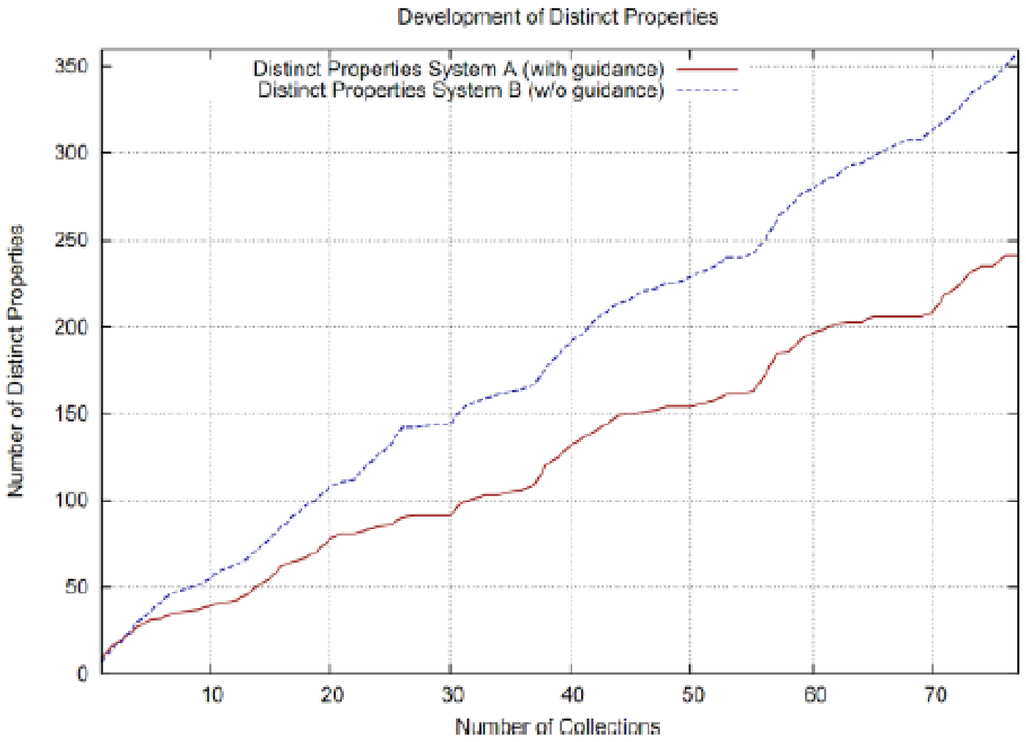

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